EviSeq v2: Evidence-Guided Cross-Attention for Composing Pretrained Language Models

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Abstract

Abstractive summarization must retain the central information in a document while producing fluent text. This interface is learned jointly in standard encoder-decoder models, but useful pretrained checkpoints are often available only as an embedding encoder and a causal decoder. We introduce EviSeq v2, a single-memory composition of PPLX-Embed and Qwen3 for source-grounded summarization. The model projects encoder states into the decoder space, pools them into sentence units, and adds a length-normalized evidence prior to copied cross-attention. DualBridge makes the prior conditional on the instruction and the source document before generation begins. An attention-aligned contrastive objective trains the unit scores that the decoder consumes at inference. The decoding graph remains one encoder, one bridge, one decoder, and greedy argmax. On PubMed, the measured PPLX composition obtains ROUGE-1 49.228, ROUGE-2 21.644, and ROUGE-L 45.312 with a Python diagnostic scorer, compared with 49.580, 21.990, and 45.463 for T5Gemma2-1B-1B. The locked DualBridge run and matched ablations are still pending Perl ROUGE-1.5.5 scoring. These measurements document the composition and define the evidence interface tested by the pending ablations.

1 Introduction

Abstractive summarization asks a model to compress a document into a short text that preserves its central information. A useful summary must be fluent, relevant, and supported by the source. Fluency alone is insufficient. A sentence can be grammatical while changing a number, omitting a qualification, or introducing a fact that is absent from the document. Attention-based sequence-to-sequence models address this problem by separating source encoding from target generation and by exposing the source at every target step (Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017).

The checkpoint ecosystem makes this interface difficult to reproduce. Strong models are commonly released as decoder-only language models or as embedding encoders rather than as a matched encoder-decoder pair (Brown et al., 2020; Dubey et al., 2024; Yang et al., 2024; Zhang et al., 2025b). The two components may encode similar content in different coordinates, and the decoder has no learned source memory for the new encoder. A direct projection can make the dimensions compatible, but it does not specify which source units should receive attention during generation. This mismatch is especially costly for long documents, where a fluent decoder can spend its cross-attention budget on locally similar but nonessential text.

Jointly pretrained systems such as T5, BART, and mT5 learn a source-to-target interface before fine-tuning (Raffel et al., 2020; Lewis et al., 2020; Xue et al., 2021). T5Gemma2 further shows that modern language-model components can be arranged as an encoder-decoder system (Zhang et al., 2025a). These models provide a strong reference, but they do not answer whether independently pretrained, smaller checkpoints can be composed with an explicit, inspectable evidence interface. The missing component is not another decoder. It is a training-compatible bridge that makes source representations readable and gives the decoder a useful prior over source units.

We propose EviSeq v2, a single-memory encoder-decoder architecture that composes PPLX-Embed with Qwen3. The key ideas are: (i) an identity-initialized residual projection and a length-normalized unit prior, which address coordinate mismatch and sentence-length bias; (ii) DualBridge, which adds a target-free prompt-conditioned route to the static source salience route; and (iii) an attention-aligned evidence contrastive objective, which trains the same unit energies that become the decoder's cross-attention bias. The model does not create a second memory bank, use reference text at inference, or rerank sampled candidates. It performs one encoder pass, one bridge pass, and greedy decoding.

The design leads to four empirical questions.

- **RQ1:** Can a PPLX-Embed encoder and a Qwen3 decoder form a useful summarizer through one evidence bridge?
- **RQ2:** Does a prompt-conditioned route improve the static source salience route without changing greedy inference?
- **RQ3:** Which part of the evidence objective affects generation quality: unit salience, representation contrast, or their direct coupling to cross-attention?
- **RQ4:** How do accuracy, length, repetition, and computational cost change across PubMed and the auxiliary WikiLingua setting?

We evaluate the composition on PubMed and use WikiLingua as an auxiliary cross-domain setting. The main comparison uses the same greedy decoder protocol for EviSeq and T5Gemma2. The available PubMed measurement places the PPLX composition close to, but below, the reference. The locked DualBridge result and matched ablations are still pending Perl ROUGE-1.5.5 scoring, so we separate measured observations from the hypotheses tested by the new interface.

Our contributions are threefold. First, we give a reproducible recipe for connecting an embedding encoder to a pretrained causal decoder through one source memory. Second, we define a target-free evidence bridge whose static and prompt-conditioned routes affect the same cross-attention prior. Third, we provide an evaluation protocol that links evidence diagnostics to generation quality while controlling decoding, output length, and scorer choice.

2 Related Work

Native encoder-decoder models. Sequence-to-sequence learning separates reading from writing, while attention exposes the source at every target step (Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017). T5, BART, and mT5 learn this interface during pretraining and transfer it to summarization (Raffel et al., 2020; Lewis et al., 2020; Xue et al., 2021). ROUGE measures lexical overlap, whereas BERTScore uses contextual similarity; neither metric alone captures factual consistency (Lin, 2004; Zhang et al., 2020; Maynez et al., 2020; Fabbri et al., 2021). EviSeq keeps these metrics for evaluation but studies how to create the source interface when the two checkpoints were not jointly pretrained.

Language models as encoders. Decoder-only checkpoints can provide useful sentence representations after representation-oriented adaptation. LLM2Vec and Generative Representational Instruction Tuning improve this use case through additional objectives (BehnamGhader et al., 2024; Muennighoff et al., 2024), and LaMaTE places a language-model encoder in an encoder-decoder translation system (Luo et al., 2025). Qwen3 Embedding provides a related embedding-oriented family (Zhang et al., 2025b). These works motivate reusing pretrained encoders, but they do not define an inspectable unit-level evidence interface for a new causal decoder.

Contrastive learning and model composition. Supervised contrastive learning separates examples that share a label from alternatives, and it can be combined with cross-entropy for language-model fine-tuning (Khosla et al., 2020; Gunel et al., 2021). SimCLS uses candidate ranking to reduce the gap between sequence likelihood and summary quality (Laban et al., 2021); EviSeq instead contrasts source units before generation and trains the prior consumed by cross-attention. T5Gemma2 is the closest composition reference (Zhang et al., 2025a). The remaining gap is to connect independently pretrained checkpoints while making evidence selection conditional, differentiable, and usable in a single greedy decoding pass.

3 Methodology

Task and Notation. Let $x = (x_1, \ldots, x_S)$ be a source document and let $y = (y_1, \ldots, y_T)$ be its reference summary. The encoder produces hidden states $H = E_\phi(x) \in \mathbb{R}^{S\times d_e}$. The decoder has hidden size d_d. Source units are sentences or sentence-like spans. A unit map $u(s) \in \{0, 1, \ldots, U\}$ assigns each source token to a unit, where zero denotes a neutral token such as a separator or padding position. The model produces one summary autoregressively with greedy decoding.

Our PubMed configuration uses `perplexity-ai/pplx-embed-v1-0.6b` as E_ϕ and `Qwen/Qwen3-0.6B` as the causal decoder. Both backbones are loaded from their pretrained checkpoints. The bridge and copied cross-attention modules are initialized so that the source route starts close to a conventional decoder interface.

The construction follows the generation path. We first correct the encoder coordinates and turn token states into unit-level evidence scores. We then condition those scores on the instruction and document context, align the resulting energies with the decoder's source attention, and finally use the same bridge at greedy inference. This order separates the interface design from the auxiliary supervision used to train it.

Design Principles and Theoretical Properties. The bridge is an attention prior, not a replacement for cross-attention. Following the energy-based view of neural attention (Bahdanau et al., 2015; Vaswani et al., 2017), let q_t be a decoder query at target position t, and let k_s be the key of source token s. The copied attention layer assigns source probability

$$A_{t,s} = \text{softmax}_s\left(\frac{q_t^\top k_s}{\sqrt{d_k}} + b_s\right). \tag{1}$$

The first term is content matching and the second term is the bridge prior. For a source unit i, define its attention mass as $\pi_{t,i} = \sum_{s:u(s)=i} A_{t,s}$. The bridge therefore tilts the decoder's native retrieval distribution while content matching can still override a weak prior.

The length correction has a simple probabilistic interpretation. Suppose the content scores are approximately constant within a unit, so that $q_t^\top k_s/\sqrt{d_k} \approx z_{t,i}$ for s in unit i. Without a correction, the unnormalized unit mass is $n_i \exp(z_{t,i}+a_i)$ and grows with the number of visible tokens n_i. With the bridge bias $a_i - \log n_i$, it becomes

$$n_i \exp(z_{t,i} + a_i - \log n_i) = \exp(z_{t,i} + a_i). \tag{2}$$

Thus, under this approximation, the prior compares units rather than token counts. Equation 2 is not an assumption about all decoder queries. It explains why the correction removes a known source of length bias without imposing a hard extractive mask.

The projection has a second role. Keys and values are computed by decoder parameters, but the encoder state H was learned in an independent coordinate system. If $P^\star$ is a linear coordinate correction that makes the source states useful to those decoder projections, then the bridge can represent $P^\star H$ while the decoder remains unchanged. When the widths match, EviSeq parameterizes $P = I + \Delta_\psi$ and initializes $\Delta_\psi = 0$. A first-order expansion around this initialization gives

$$\mathcal{L}(PH) \approx \mathcal{L}(H) + \langle \nabla_M \mathcal{L}(H), \Delta_\psi H \rangle. \tag{3}$$

The initialization therefore preserves the pretrained source route at step zero, while the gradient can learn a task-specific rotation, scale, or low-rank correction. It does not guarantee that the two representations are globally aligned. The projection residual and the held-out generation score are the empirical checks for that claim.

DualBridge makes the prior conditional on the task and document. A static score has the form $a_i = f(M_i)$. The dynamic route instead computes $d_i = g(M_i, h_0(p), c_x)$, where $h_0(p)$ is the decoder state induced by the fixed instruction and c_x is pooled source context. The fused score $\tilde{a}_i = a_i + \sigma(r)d_i$ therefore approximates a conditional unit preference $s(i \mid x, p)$ rather than a source-only preference. Because h_0 is obtained before the first target token, this conditioning is available at inference without reference text or teacher forcing.

Finally, the evidence loss follows supervised contrastive training (Khosla et al., 2020; Gunel et al., 2021) and is aligned with the quantity consumed by Equation 1. For one positive unit p, let $r_p = \log \sum_{n \in N_H} \exp \ell_n - \ell_p$ and $\eta_p = \sigma(r_p)$. For an unclipped prior energy e_i, the derivatives of the per-positive softplus term are

$$\begin{aligned} \frac{\partial \mathcal{L}_p}{\partial e_p} &= -\frac{\beta}{\tau}\eta_p, \\ \frac{\partial \mathcal{L}_p}{\partial e_n} &= \frac{\beta}{\tau}\eta_p \frac{\exp \ell_n}{\sum_{m \in N_H} \exp \ell_m}. \end{aligned} \tag{4}$$

For $\beta > 0$, gradient descent increases positive prior energy and decreases hard-negative energy. If clipping is active, the corresponding derivative is zero, which is why the implementation reports the clipping fraction. This calculation distinguishes attention-aligned contrastive learning from a representation-only auxiliary loss: the latter cannot directly change the additive bias used by the decoder.

Projection and Static Evidence Bridge. The encoder and decoder do not share a guaranteed hidden-space coordinate system. A direct identity connection would preserve dimensions but would not correct a coordinate mismatch. EviSeq therefore applies an identity-initialized residual projection when the dimensions match. When dimensions differ, it uses RMS normalization followed by a linear map. Denote the projected memory by

$$M = \Pi_\psi(H) \in \mathbb{R}^{S \times d_d}. \tag{5}$$

The projection parameters are ψ. The residual initialization preserves the pretrained memory at step zero while allowing fine-tuning to learn a coordinate correction. This design keeps the bridge small relative to the two backbones and makes its effect measurable through a projection residual diagnostic.

For unit i, we mean-pool the projected token states and apply an evidence head to obtain a static salience logit a_i. The valid-unit mask excludes padded and neutral units. The bridge turns unit logits into a token-level additive bias. If token t belongs to unit i with n_i visible tokens, its bias is

$$b_t = g_s \ \text{clip}(a_i, -5, 5) - \log n_i. \tag{6}$$

Here $g_s = \sigma(\gamma_s)$ is a positive learned salience gate and n_i is the visible token count of unit i. The $-\log n_i$ term makes the total prior mass of a unit approximately invariant to its token length. Neutral tokens receive a normalized neutral bias. The resulting vector $b \in \mathbb{R}^S$ is passed to every copied decoder cross-attention layer. Content-based attention remains free to retrieve any source token; b is a prior, not a hard mask.

The decoder layer first applies its native causal self-attention. It then applies an independently copied source attention with the same query, key, value, and output dimensions as the corresponding self-attention block. A scalar cross gate controls the residual source update. At initialization, this gate is small, so the decoder is not forced to trust an unaligned source memory before the bridge has learned.

DualBridge: A Target-Free Dynamic Route. Static salience is source-only. It cannot condition its unit preference on the task instruction that precedes generation. DualBridge adds a prompt-conditioned route while retaining the static route and the single projected memory.

Let p be the fixed decoder instruction and let h_0 be the decoder state at the final prompt position. We obtain h_0 with a neutral source prior, which gives every valid source unit equal total prior mass. Only the final L_p copied cross-attention layers are active in this short probe. We set $L_p = 2$. The probe is target-free because no reference summary token is visible.

We pool the projected source memory by unit and then across valid units. The prompt and source context form a query, while each unit forms a key:

$$\begin{aligned} c_x &= \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} M_i, \\ q &= \text{norm}(W_p h_0 + \sigma(g_c) W_m c_x), \end{aligned} \tag{7}$$

$$k_i = \text{norm}(W_k M_i). \tag{8}$$

Here M_i is the mean-pooled memory of unit i, W_p, W_m, and W_k are trainable projections, and g_c controls the source-context contribution. The query is task-aware through h_0 and document-aware through c_x. The dynamic unit score is

$$\begin{aligned} d_i &= \text{clip}\Big(\alpha q^\top k_i, -d_{\text{max}}, d_{\text{max}}\Big), \\ \widetilde{a}_i &= a_i + \sigma(r) d_i. \end{aligned} \tag{9}$$

We set $\alpha = 8$, $d_{\text{max}} = 2$, and initialize the fusion gate $\sigma(r)$ to 0.50. The fused logits $\widetilde{a}_i$ replace a_i in Equation 6. Thus DualBridge adds a dynamic decision but not a second memory bank or a second decoder. The same probe and fusion are used during teacher-forced training and before greedy generation.

Figure 1 summarizes the data flow. The fixed instruction is used only to construct the target-free probe state. Reference summaries and evidence labels are used by training losses, not by the inference graph.

Evidence Labels and Hard Negatives. The prepared PubMed records contain a list of source sentences, a reference summary, and a `label` list of extractive evidence indices. The preparation script greedily adds the sentence that maximizes the sum of unigram and bigram ROUGE against the complete reference, up to three units. This creates weak extractive supervision for a model whose output remains abstractive. The selection

Source document x $\longrightarrow$ **PPLX-Embed encoder** E_ϕ $\longrightarrow$ **Identity-initialized projection** Π_ψ

M with sentence units M_i

Static unit head	DualBridge prompt probe	Evidence fusion
a_i	fixed instruction $p \to h_0$	$\widetilde{a}_i = a_i + \sigma(r) d_i$
length-normalized prior	neutral source prior, two copied layers	$d_i = \mathrm{clip}(\alpha q^\top k_i)$

$\widetilde{a}_i \longrightarrow$ token prior $b_t \longrightarrow$ copied cross-attention in Qwen3 decoder $\longrightarrow$ greedy summary

Training only: up to three evidence positives and within-document hard negatives supervise the query, keys, salience logits, and live attention energy.

Figure 1: EviSeq v2 DualBridge. The model keeps one projected source memory and one Qwen3 decoder. The prompt probe is target-free and is reused at inference. Evidence annotations affect only training objectives and diagnostics.

is performed separately within each training, validation, and test record. It is not a model prediction.

For a training document, $P(x)$ denotes its valid positive units and $N_H(x)$ denotes the mined hard negatives. A hard negative is a non-evidence unit with high detached query-key similarity. The selection also adds a small detached boost for high current salience, which prioritizes false positives. We use four hard negatives during interface warm-up and eight during full fine-tuning. If a document has fewer valid negatives, the loss skips that example rather than fabricating a negative.

The test records may retain evidence indices for diagnostics. The evaluator does not pass those labels to the greedy generation function. A test label can therefore be used to measure evidence accuracy after decoding, but it cannot change the generated text. This distinction is required because the labels were constructed with the reference summary. The next subsection uses these labels only to train the unit energy that remains available when the reference is absent.

Attention-Aligned Evidence Contrastive Learning. Let q and k_i be the normalized DualBridge query and keys. Let e_i be the live unit attention energy consumed by the bridge, including the salience gate, bias scale, and logit clipping. For each positive unit p, the evidence loss contrasts it with the same hard-negative set:

$$\mathcal{L}_{\text{evi}} = \frac{1}{|P(x)|} \sum_{p \in P(x)} \text{softplus}\Bigg(\log \sum_{n \in N_H(x)} \exp \ell_n - \ell_p \Bigg), \tag{10}$$

$$\ell_i = \frac{q^\top k_i + \beta e_i}{\tau}. \tag{11}$$

The temperature τ controls the sharpness of the comparison and is set to 0.07. The coefficient β couples the contrastive target to the live attention prior and is set to 0.15. A positive unit is encouraged to have a larger query-key score and a larger consumed prior energy than its hard negatives. The detached mining step receives no gradient. The final scores in Equation 11 do receive gradient, so the loss reaches the prompt-conditioned head, bridge projection, salience head, and attention gate.

The training objective combines generation, salience, and evidence terms:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_s \mathcal{L}_{\text{sal}} + \lambda_e \mathcal{L}_{\text{evi}}, \quad \lambda_s = 0.10, \qquad \lambda_e = 0.10. \tag{12}$$

Here $\mathcal{L}_{\text{CE}}$ is teacher-forced token cross-entropy, $\mathcal{L}_{\text{sal}}$ is balanced pointwise salience loss with an optional within-document ranking term, and $\mathcal{L}_{\text{evi}}$ is Equation 10. The auxiliary terms are scaled only during training. In evaluation, the model uses the bridge to generate text and reports the configured task metric.

Inference. Inference follows four steps. The encoder reads the source document. The bridge projects and pools the memory. DualBridge runs the fixed decoder probe with a neutral source prior and fuses the resulting dynamic scores with static salience. The Qwen3 decoder then generates one summary with greedy argmax decoding. It uses one beam, with sampling disabled. No reference summary, extractive label, candidate pool, reranker, or teacher-forced target state is used.

4 Experiments

Research Questions and Scope. RQ1 is tested by comparing the PPLX to Qwen3 composition with T5Gemma2-1B-1B on PubMed. RQ2 is tested by comparing the static bridge, PCEB, and

DualBridge under the same data, optimizer, and decoding protocol. RQ3 uses ablations that remove the evidence loss, remove its direct attention coupling, or remove the bridge prior. RQ4 measures the same systems on the auxiliary WikiLingua and CNN/DailyMail configurations. The final RQ2 to RQ4 values are TBD until the locked runs finish.

Datasets and Preprocessing. The main dataset is PubMed. Our prepared split contains 116,699 training records and 6,657 test records. The exact validation count is recorded with the data manifest and will be inserted as TBD if it differs between data snapshots. Each record contains `text` as a list of source sentences and `summary` as a list of reference sentences. The source is serialized with one sentence per line. The source limit is 4,096 tokens and the target limit is 512 tokens.

WikiLingua is an auxiliary cross-domain setting with 13,999 training records, 1,680 validation records, and 3,901 test records in our data snapshot. It is smaller and more variable than PubMed. We use it to test whether the interface behaves outside the main biomedical domain, not to select the PubMed checkpoint. CNN/DailyMail is configured as a further English test, but its final EviSeq v2 result is not available in this draft.

The evidence labels are prepared before training. The greedy selector uses whole-document reference overlap to identify up to three source units. During training, these labels supervise salience and evidence contrast. During test generation, they are masked from the generation function. The preparation pipeline checks duplicate identifiers and cross-split source content before training.

Models and Training. The main composition uses PPLX-Embed v1 0.6B as the encoder and Qwen3-0.6B as the decoder. The bridge uses an identity-initialized residual projection, a 512-dimensional salience hidden layer, unit-invariant length normalization, and a sigmoid salience gate initialized at 0.10. The decoder inserts copied cross-attention at every layer and initializes the cross gate at 0.10. DualBridge uses two copied cross-attention layers for the neutral prompt probe, a dynamic logit scale of 8, a dynamic clip of 2, and a prompt route mixture initialized at 0.50.

The main schedule has one interface warm-up epoch and four full-fine-tuning epochs. Warm-up aligns the copied cross-attention and bridge while keeping the backbone update restricted. Full fine-tuning updates the pretrained encoder and decoder together with the bridge and evidence heads. The EviSeq v2 recipe does not use LoRA. Each backbone is nominally a 0.6B checkpoint; the exact footprint including copied attention and bridge modules is reported by the runtime parameter manifest because checkpoint revisions can change rounded totals.

The PubMed recipe uses a physical batch size of 32 and gradient accumulation of 1 in the locked model configuration. Training uses gradient checkpointing, fused optimization when available, BF16 on the server, and a fixed seed of 42. Validation selects `best.pt` by cross-entropy. We save `last.pt` and each epoch checkpoint for recovery. These choices are implementation details only when they affect reproducibility or memory use, so we do not interpret them as architectural contributions.

Decoding and Metrics. All systems use the same decoder instruction, source and target limits, checkpoint rule, and greedy generation. We set `do_sample=false` and `num_beams=1`. The main automatic metric is ROUGE-1, ROUGE-2, and ROUGE-L F1. The diagnostic scorer is a Python ROUGE implementation. The paper scorer is Perl ROUGE-1.5.5 through pyrouge with the configured stemming and sentence tokenization. A score from one backend is not substituted for a score from the other backend.

We also report mean prediction length, mean reference length, length ratio, empty-prediction rate, too-short and too-long rates, repeated-trigram rate, and unique prediction rate. These diagnostics test whether a metric change is explained by output length or repetition. Evidence top-1 accuracy and the positive to hard-negative similarity gap measure the auxiliary task. The positive attention-prior gap measures the difference in the live bridge energy between labeled positives and negatives. Cross residual ratio and projection residual ratio measure whether the source route is active and whether the coordinate correction remains bounded.

For the final comparison, each model is evaluated once on the locked test split. Paired bootstrap confidence intervals will be computed over identical examples with the same Perl scorer. Statistical significance is not claimed for the provisional numbers below because they do not include a paired uncertainty estimate.

System	R-1	R-2	R-L
T5Gemma2-1B-1B	49.580	21.990	45.463
EviSeq v2, PPLX, measured run	49.228	21.644	45.312
EviSeq v2, PPLX, second measured run	48.976	21.482	45.078
EviSeq v2, Qwen3 Embedding	48.439	20.888	44.594
EviSeq v2 DualBridge, locked run	TBD	TBD	TBD

Table 1: Provisional PubMed diagnostic ROUGE F1 percentages; DualBridge is unscored.

5 Results and Discussion

PubMed Main Comparison. Table 1 reports the PubMed measurements currently available. The T5Gemma2 row is the reference run. The EviSeq rows are measured PPLX and Qwen3-Embedding compositions. The final DualBridge configuration is the method described in Section 3; its locked test score is not yet available. The table therefore separates a measured checkpoint from the result required to answer RQ2.

The measured PPLX composition is close to the T5Gemma2 reference but does not exceed it in any listed metric. Its gap is 0.352 ROUGE-1 points, 0.346 ROUGE-2 points, and 0.151 ROUGE-L points. The Qwen3 Embedding composition is lower on this snapshot. These observations answer only the narrow question of the measured checkpoints. They do not determine whether the final DualBridge route helps, because its test row remains TBD.

The scores also illustrate why the evaluation backend must be locked. Earlier experiment outputs used both Python ROUGE and Perl ROUGE-1.5.5. Perl scoring can differ because of tokenization, stemming, and the Perl implementation. We will not compare the provisional rows to the T5Gemma2 row as if they came from different backends. The final table will replace the provisional label after all predictions are scored by the same Perl command.

Training Diagnostics. The runtime diagnostics provide evidence that the auxiliary path is connected, but they are not substitutes for test ROUGE. In an observed interface warm-up trace, the evidence contrastive loss was approximately 1.63, the evidence weight was about 0.13, the evidence accuracy ranged from 0.17 to 0.25, and the similarity gap was positive but small. The cross residual ratio was about 0.036, and the bridge projection residual ratio was between 0.24 and 0.26. These values are consistent with a source route that is active while still being early in optimization.

A later validation diagnostic recorded evidence accuracy near 0.55, a positive similarity gap near 0.146, evidence contrastive loss near 1.388, and a cross residual ratio near 0.0348. The attention-prior gap was positive. These values show that the query and key representations separate labeled units in that checkpoint. They do not establish that the separation improves summary quality. The ablation and locked test rows are required for that conclusion.

Ablations and Secondary Datasets. We pre-register four matched PubMed ablations: CE plus salience, static evidence contrast, PCEB without the dynamic inference route, and full DualBridge. Their scores are TBD. Each run keeps the source serialization, optimizer, seed, decoding, training passes, and validation checkpoint rule fixed, and will be scored on the same examples with Perl ROUGE-1.5.5.

Earlier WikiLingua and CNN/DailyMail measurements are excluded until the DualBridge recipe is scored with the same protocol.

6 Analysis

The theory predicts two observable effects. The residual projection should reduce the encoder-decoder coordinate mismatch while remaining close to its identity initialization, and the length-normalized prior should change unit odds without preventing content-based retrieval. The prompt route should make that prior depend on both the instruction and the document. We test these predictions with projection and cross-residual diagnostics, attention-prior gaps, matched ablations, and locked greedy ROUGE.

The measured PPLX composition is close to, but below, T5Gemma2 on the provisional PubMed metrics. A positive evidence gap and active cross residual show that the auxiliary path is connected, not that it improves generation. The lower Qwen3 Embedding result is descriptive because the encoder variants were not compared under a new matched Perl-scored run. We reject an improvement claim if it is explained by output length, repetition, or a different scorer. A positive evidence gap without a ROUGE gain would indicate proxy misalignment.

7 Limitations and Responsible Use

The main evidence comes from one biomedical dataset and one encoder-decoder composition, so it does not support a claim of robustness across domains or model families. The extractive labels are weak reference-overlap supervision, not human factuality annotations. ROUGE can penalize valid paraphrases and does not measure factual consistency by itself. The recipe full-fine-tunes both backbones, and adapter comparisons require a separate controlled budget.

8 Conclusion

EviSeq v2 composes a PPLX-Embed encoder and a Qwen3 causal decoder through one projected source memory and one evidence bridge. DualBridge adds a prompt-conditioned, target-free route that modifies the same unit prior consumed by decoder cross-attention. Attention-aligned contrastive learning supervises the query, source keys, salience logits, and live bridge energy while keeping greedy inference unchanged.

The measured PubMed composition answers RQ1 positively in a limited sense: the independently pretrained checkpoints form a usable summarizer, although the observed score remains below T5Gemma2. RQ2 to RQ4 require the locked DualBridge test result and matched ablations. The main takeaway is therefore an inspectable, reproducible interface whose superiority must be supported by controlled experiments rather than inferred from auxiliary diagnostics.

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A Additional Derivations

This appendix gives the derivations behind the two interface properties used in the main paper. They are included for completeness; the main claims depend on the matched ablations and locked test scores.

A.1 Unit-Length Invariance

Let $\mathcal{S}_i = \{s : u(s) = i\}$ contain n_i visible tokens. The unnormalized cross-attention mass of unit i at target position t is

$$W_{t,i} = \sum_{s \in \mathcal{S}_i} \exp\big(z_{t,s} + g_s \,\mathrm{clip}(a_i, -5, 5) - \log n_i\big). \tag{13}$$

where $z_{t,s} = q_t^\top k_s / \sqrt{d_k}$. If $z_{t,s} = z_{t,i}$ within the unit, then

$$W_{t,i} = \exp(z_{t,i} + g_s \,\mathrm{clip}(a_i, -5, 5)) \,. \tag{14}$$

The unit length no longer appears in the relative mass. When token content differs within a unit, the correction still removes the multiplicative count factor and leaves the residual variation to content attention. The result is a prior over units, not a sentence selector.

A.2 Gradient of the Attention-Aligned Loss

For a positive unit p and hard-negative set N_H, define $r_p = \log \sum_{n \in N_H} \exp \ell_n - \ell_p$ and $L_p = \log(1 + \exp r_p)$. Since $\partial L_p / \partial r_p = \sigma(r_p)$, the chain rule gives

$$\begin{aligned} \frac{\partial L_p}{\partial \ell_p} &= -\sigma(r_p), \\ \frac{\partial L_p}{\partial \ell_n} &= \sigma(r_p) \,\frac{\exp(\ell_n)}{\sum_{m \in N_H} \exp(\ell_m)}. \end{aligned} \tag{15}$$

With $\ell_i = (q^\top k_i + \beta e_i)/\tau$, the derivatives with respect to the live bridge energy are Equation 4. For $\beta > 0$, a gradient step raises the positive energy and lowers the weighted hard-negative energies. In the implementation $e_i = g_s \,\mathrm{clip}(a_i, -5, 5)$ up to the configured bias scale. A unit at the clipping boundary receives zero derivative through the clip, so the clipping fraction is logged as a saturation check.

A.3 Conditional Routing

The static route is $a_i = f(M_i)$. DualBridge adds $d_i = g(M_i, h_0(p), c_x)$ and uses $\widetilde{a}_i = a_i + \sigma(r) d_i$. If two documents have different pooled contexts, $c_x^{(1)} \neq c_x^{(2)}$, the same instruction can produce different queries even when the prompt tokens are identical. If the dynamic gate is zero, the model reduces exactly to the static bridge. This gives a direct nested ablation: the dynamic route can be removed without changing the memory, decoder, or static salience head.

B Configuration and Reproducibility

Table 2 records the configuration used by the PubMed EviSeq v2 DualBridge recipe. Values marked as provisional in the main paper remain provisional here.

The encoder and decoder are fine-tuned jointly during the full stage. The evidence labels, hard-negative identities, and validation checkpoint are recomputed or loaded from the same prepared data snapshot for each matched run. No model is downloaded by the evaluation script, and no reference text is passed to generation.

C Evidence Construction and Evaluation Protocol

For each record, the preparation script treats the source as an ordered list of sentences and the abstract as an ordered list of reference sentences. It greedily adds the unused source sentence that gives the largest increase in the sum of unigram and bigram ROUGE against the complete reference, up to three units. The resulting indices are stored in the record's `label` field. The procedure is an oracle-style training signal, not a prediction made by EviSeq.

The same field may remain in validation and test files for diagnostics. The evaluator masks it before greedy generation. In particular, the test label cannot modify the prompt probe, the fused prior, the decoder cache, or the generated token distribution. A test label is used only after generation to compute evidence accuracy or attention-prior gaps.

The primary metric is Perl ROUGE-1.5.5 through pyrouge with a fixed file naming scheme, stemming option, and sentence tokenization. The Python ROUGE implementation is retained as a diagnostic backend and is not mixed with Perl scores in the comparison table. All systems use the same

Component	Configuration
Encoder	PPLX-Embed v1 0.6B, `perplexity-ai/pplx-embed-v1-0.6b`
Decoder	Qwen3-0.6B, `Qwen/Qwen3-0.6B`
Source and target limits	4,096 and 512 tokens
Memory	One projected source memory, sentence units
Projection	Identity-initialized residual map when widths match
Static bridge	512 hidden units, sigmoid gate 0.10, unit-invariant length correction
Copied cross-attention	Inserted in every decoder layer, cross gate 0.10
Prompt probe	Two final copied layers, neutral source prior
Dynamic route	Scale 8, clip 2, fusion gate 0.50
Evidence contrast	Temperature 0.07, attention coupling 0.15, weight 0.10
Hard negatives	Four in warm-up, eight in full fine-tuning
Evidence positives	Up to three global source units per document
Optimization	One warm-up epoch, four full-fine-tuning epochs, BF16, seed 42
Batching	Physical batch 32, gradient accumulation 1
Decoding	Greedy argmax, one beam, sampling disabled
Checkpoint rule	Best by validation CE, with last and per-epoch checkpoints

Table 2: Current PubMed DualBridge configuration.

source serialization, decoder instruction, maximum lengths, seed, checkpoint rule, and greedy decoder settings. Paired uncertainty will be computed on identical test records after the final predictions are available.

This paper reports the measured PubMed rows supplied by the experiment log and labels the final DualBridge and ablation rows as TBD. This appendix does not turn an unmeasured row into evidence. Its purpose is to make the intended experiment and the information available at each stage explicit.